ML 2026 • Lecture 11, reading

The support vector machine, step by step: from "which line?" to the dual

Companion to the Lecture 11 slides. Same eight points, same numbers. Read it once before the class and once after. The explorer is at `https://ml2026.floodresq.com/explore/svm`.

1. The question

Eight emails, two features each (x_1, the number of links; x_2, the number of words written in capitals), and a label $y \in \{+1, -1\}$: $+1$ spam, -1 not spam (a normal email).

i	1	2	3	4	5	6	7	8
x_1	3	4	3	4	1	0	1	0
x_2	3	3	4	4	1	1	0	0
y	$+1$	$+1$	$+1$	$+1$	-1	-1	-1	-1

Many straight lines separate the two groups without a single mistake. On the training data they are all equally good, so "fewest errors" cannot choose. We need a rule that does. The rule of this lecture:

The rule. Among all lines that separate the classes, choose the one that stays *farthest from the nearest data point*. A line that grazes a point is one small nudge away from an error; a line with a wide empty corridor on both sides tolerates noise and new data.

Everything that follows turns this sentence into mathematics and then solves it. Three things must be made precise: what "distance from a point to a line" is, what "nearest" means when points sit on two sides, and what we are free to choose.

2. A line is a direction w and an offset b

Write a line as the set of points $\mathbf{x}$ with $\mathbf{w}^\top \mathbf{x} + b = 0$. For the line $x_1 + x_2 = 4$ this is $\mathbf{w} = (1, 1)$, $b = -4$.

$\mathbf{w}$ is *not a point on the line*; it is a direction. Take two points on the line, $\mathbf{p} = (0.5, 3.5)$ and $\mathbf{q} = (3.5, 0.5)$. Both satisfy the equation, so subtracting the two equations gives $\mathbf{w}^\top(\mathbf{q} - \mathbf{p}) = 0$: check, $(1, 1) \cdot (3, -3) = 0$. The vector $\mathbf{q} - \mathbf{p}$ is a step *along* the line, and a zero dot product means $\mathbf{w}$ is perpendicular to it. So $\mathbf{w}$ points *across* the line.

The number $s(\mathbf{x}) = \mathbf{w}^\top \mathbf{x} + b$ is the **score** of a point: zero on the line, positive on the side $\mathbf{w}$ points to, negative on the other side. For $x_1 + x_2 - 4$: the score of $(3, 3)$ is $+2$, of $(1, 1)$ is -2, of $(2, 2)$ is 0.

3. Distance from a point to the line

Start at $\mathbf{x}$ and walk straight toward the line, i.e. along $-\mathbf{w}/\|\mathbf{w}\|$, the unit vector across it. After a distance t you stand at $\mathbf{x} - t\,\mathbf{w}/\|\mathbf{w}\|$. You are on the line when the score there is zero:

$$\mathbf{w}^\top\Big(\mathbf{x} - t\frac{\mathbf{w}}{\|\mathbf{w}\|}\Big) + b = 0 \quad\Longrightarrow\quad \mathbf{w}^\top\mathbf{x} + b - t\,\frac{\mathbf{w}^\top\mathbf{w}}{\|\mathbf{w}\|} = 0 \quad\Longrightarrow\quad \boxed{d(\mathbf{x}) = t = \frac{\mathbf{w}^\top\mathbf{x} + b}{\|\mathbf{w}\|}}$$

using $\mathbf{w}^\top\mathbf{w} = \|\mathbf{w}\|^2$. **Distance is score divided by the length of w.** Its sign tells the side.

Worked: $\mathbf{x} = (4, 3)$ and the line $x_1 + x_2 = 4$. Score $4 + 3 - 4 = 3$; $\|\mathbf{w}\| = \sqrt{1^2 + 1^2} = 1.414$; $d = 3/1.414 = 2.12$. The foot of the perpendicular is $(4, 3) - 2.12\,(0.707, 0.707) = (2.5, 1.5)$, and indeed $2.5 + 1.5 = 4$. For $(1, 1)$: score -2, $d = -1.41$, the other side.

One fact to keep: multiply $\mathbf{w}$ and b by the same number and the distance does not change. Scale by 3: score 9, $\|\mathbf{w}\| = 4.24$, ratio still 2.12. Distance belongs to the *line*, not to the pair $(\mathbf{w}, b)$.

4. The margin: the distance to the nearest point

A spam point should have $d > 0$, a non-spam point $d < 0$. Multiply by the label: $y_i\, d(\mathbf{x}_i)$ is positive exactly when point i is on its correct side. For the line $x_1 + x_2 = 4$:

i	1	2	3	4	5	6	7	8
score	2	3	3	4	−2	−3	−3	−4
$d = \text{score}/1.414$	1.41	2.12	2.12	2.83	−1.41	−2.12	−2.12	−2.83
$y_i\, d(\mathbf{x}_i)$	**1.41**	2.12	2.12	2.83	**1.41**	2.12	2.12	2.83

$$\text{margin}(\mathbf{w}, b) = \min_i \; y_i \, \frac{\mathbf{w}^\top \mathbf{x}_i + b}{\|\mathbf{w}\|}.$$

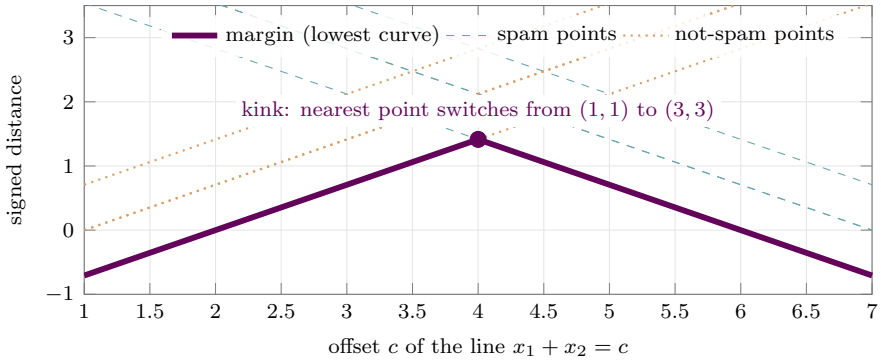
For $x_1 + x_2 = 4$ it is 1.414, attained by points 1 and 5. Those two points touch the edges of the corridor.

5. The rule as a formula, and why it cannot be attacked directly

$$\max_{\mathbf{w},b} \Big[\min_i \; y_i \, \frac{\mathbf{w}^\top \mathbf{x}_i + b}{\|\mathbf{w}\|} \Big].$$

Inner min: for a given line, find its nearest point. Outer max: over all lines, pick the one whose nearest point is farthest. The data are fixed numbers; the unknowns are $\mathbf{w}$ and b.

This formula is the whole idea. It is also the wrong shape for a solver, and the inner min is the culprit. Each point on its own gives a smooth function of the line. The min glues the eight together and *switches* from one to another wherever two points tie for nearest. At a switch the graph has a sharp corner, a **kink**. The picture below slides the line $x_1 + x_2 = c$ and plots each point's distance against c: eight straight lines, and the margin is the lowest one at each c.



The best line, $c = 4$, sits exactly on the kink, where two points tie. There the slope jumps from $+0.707$ to -0.707 and is never zero, so "set the derivative to zero" finds nothing, and a gradient climber hops back and forth across the ridge. With the full set of lines (any direction, any offset) the same happens on a surface: creases wherever two points tie, and the maximum on a crease. Two clean-up steps fix this without changing the answer.

6. Clean-up step 1: one line has many $(\mathbf{w}, b)$; we choose the scale

$(1, 1), -4$ and $(\frac{1}{2}, \frac{1}{2}), -2$ and $(\frac{1}{4}, \frac{1}{4}), -1$ are the same line, and Section 3 showed the distance does not care which we use. So we may impose one rule that picks one $(\mathbf{w}, b)$ per line. The rule:

Scale $(\mathbf{w}, b)$ so that the nearest point scores exactly 1: $\min_i \; y_i(\mathbf{w}^\top \mathbf{x}_i + b) = 1$. Then $\text{margin} = \dfrac{\text{nearest score}}{\|\mathbf{w}\|} = \dfrac{1}{\|\mathbf{w}\|}$.

$(\mathbf{w}, b)$	$\|\mathbf{w}\|$	nearest score	margin
$(1, 1), -4$	1.414	2	$2/1.414 = 1.414$
$(\frac{1}{2}, \frac{1}{2}), -2$	0.707	**1**	$1/0.707 = 1.414$
$(\frac{1}{4}, \frac{1}{4}), -1$	0.354	0.5	$0.5/0.354 = 1.414$

Same line, same margin, three scales; only the middle row obeys the rule. A second line, $2x_1+x_2 = 5.5$: its nearest score is 2.5 (point 5), so divide by 2.5 to get $(0.8, 0.4), -2.2$ with $\|\mathbf{w}\| = 0.894$ and margin $1/0.894 = 1.118$. The messy inner min has become the single number $1/\|\mathbf{w}\|$, and comparing lines is comparing $\|\mathbf{w}\|$: 0.707 beats 0.894. **Wide margin means short w.**

7. Clean-up step 2: the problem we actually solve

Maximising $1/\|\mathbf{w}\|$ is minimising $\|\mathbf{w}\|$, which is minimising $\frac{1}{2}\|\mathbf{w}\|^2$ (same minimiser; no square root; the $\frac{1}{2}$ cancels a 2 when we differentiate). And "the nearest score is 1" can be written as "every score is at least 1": the nearest points then score exactly 1 and the others more. Nothing is lost, because if all scores were above 1, say the smallest were 1.3, dividing $\mathbf{w}$ and b by 1.3 would keep every score ≥ 1 and shorten $\mathbf{w}$, so the minimiser always has some point at exactly 1.

The primal problem (hard margin). $\min_{\mathbf{w},b} \ \frac{1}{2}\|\mathbf{w}\|^2$ subject to $y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1$ for every point i.

Objective: "make the corridor $1/\|\mathbf{w}\|$ wide". Constraint i: "point i on its correct side, outside the corridor".

Names: this is the *primal* because it is written in the line's own unknowns $(\mathbf{w}, b)$; the *dual*, in other unknowns, comes in Section 9. *Hard margin* means no point may be inside the corridor at all.

Shape: a quadratic bowl minimised over a region cut out by n half-spaces. The bowl $\frac{1}{2}\|\mathbf{w}\|^2$ is strictly convex (perfectly round), so the best $\mathbf{w}$ is unique, and the active constraints then fix b.

With the numbers in, there are three unknowns and eight constraints:

$$\begin{array}{cccc} 3w_1+3w_2+b \geq 1 & 4w_1+3w_2+b \geq 1 & 3w_1+4w_2+b \geq 1 & 4w_1+4w_2+b \geq 1 \\ -(w_1+w_2+b) \geq 1 & -(w_2+b) \geq 1 & -(w_1+b) \geq 1 & -b \geq 1 \end{array}$$

By hand: add the first and the fifth: $2w_1+2w_2 \geq 2$, so $w_1+w_2 \geq 1$. The shortest $\mathbf{w}$ with $w_1+w_2 \geq 1$ is $(\frac{1}{2}, \frac{1}{2})$; then the fifth constraint gives $b \leq -2$ and the first $b \geq -2$, so $b = -2$. Check all eight scores: $1, 1.5, 1.5, 2,\ 1, 1.5, 1.5, 2$, all ≥ 1. Margin $1/\|\mathbf{w}\| = 1.414$. Two constraints (points 1 and 5) hold with equality and decided everything; six have room to spare. We will now learn to find those two without guessing, and to see why only they matter.

8. Trade the constraints for penalties at a fixed rate

The constraints are what make the primal hard: "set the derivative to zero" knows nothing about a constraint. The way out is to charge for breaking a constraint instead of forbidding it.

To see the idea with nothing in the way, take the smallest problem of the same kind:

$$\text{(E)} \qquad \min_w \ \tfrac{1}{2}w^2 \quad \text{subject to} \quad w \geq 1, \qquad \text{answer } w = 1, \text{ value } p^* = \tfrac{1}{2}.$$

Step 1: fix a rate. Pick a number $\alpha \geq 0$ and write

$$\mathcal{L}(w, \alpha) = \tfrac{1}{2}w^2 - \alpha\,(w-1).$$

There is no constraint any more. The term $-\alpha(w-1)$ charges α per unit that w falls short of 1 and refunds α per unit that w exceeds 1.

Step 2: minimise by calculus. $\partial\mathcal{L}/\partial w = w - \alpha = 0$, so $w = \alpha$, and the minimum value is

$$D(\alpha) = \min_w \mathcal{L}(w, \alpha) = \tfrac{1}{2}\alpha^2 - \alpha(\alpha-1) = \alpha - \tfrac{1}{2}\alpha^2.$$

D is called the **dual function**: for each rate, the least penalised cost.

Step 3: $D(\alpha)$ never exceeds the true answer. $D(\alpha)$ is the smallest value of $\mathcal{L}(w, \alpha)$ as w runs over every number: imagine listing $\mathcal{L}(w, \alpha)$ for every w and taking the smallest entry. The value $w = 1$ is one entry on that list, and the smallest entry cannot be bigger than any single entry, so

$D(\alpha) \le \mathcal{L}(1,\alpha)$. At $w = 1$ the penalty $-\alpha(w-1)$ is zero, so $\mathcal{L}(1,\alpha) = \frac{1}{2}\cdot 1^2 = \frac{1}{2} = p^*$. Hence $D(\alpha) \le p^*$ for every $\alpha \ge 0$.

With $\alpha = 0.5$, a few entries on the list: $\mathcal{L}(0, 0.5) = 0.5$, $\mathcal{L}(0.5, 0.5) = 0.375$, $\mathcal{L}(1, 0.5) = 0.5$, $\mathcal{L}(1.5, 0.5) = 0.625$, $\mathcal{L}(2, 0.5) = 1$. The smallest is 0.375, and $0.375 \le 0.5$. The only reason the entry at $w = 1$ matters is that it equals the true answer, because the penalty vanishes there.

(The same argument works for any constrained problem: at the true answer every constraint is satisfied, so the penalty is zero or a refund and $\mathcal{L}$ there is at most the true cost; the smallest entry is smaller still.)

Step 4: different rates, different bounds.

rate α	minimiser $w = \alpha$	$D(\alpha) = \alpha - \frac{1}{2}\alpha^2$	is that w allowed?	gap $p^* - D(\alpha)$
0.5	0.5	0.375	no: $0.5 < 1$	0.125
1	**1**	**0.5**	yes, and it is the answer	**0**
1.5	1.5	0.375	yes, but not the shortest	0.125
2	2	0	yes, worse still	0.5

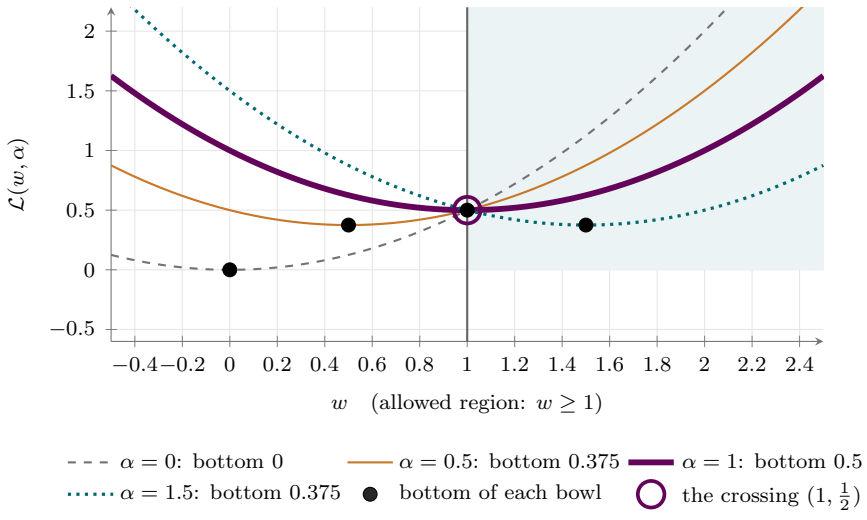
A rate that is too small lets the minimiser cheat by going below 1; a rate that is too large pushes it needlessly past 1. Only at the right rate is the unconstrained minimiser the constrained answer, and that is also the rate where the bound is tight.

Step 5: choose the rate that gives the largest bound. Maximise $D(\alpha) = \alpha - \frac{1}{2}\alpha^2$: $1 - \alpha = 0$, so $\alpha = 1$, $D = \frac{1}{2} = p^*$.

This is the dual problem: $d^* = \max_{\alpha \ge 0} D(\alpha)$. It is a maximisation because we are choosing, among a family of lower bounds, the largest one. Nothing about the original goal changed; the maximisation is Step 5.

9. Duality: the two values, the gap, and when it closes

Draw $\mathcal{L}(w,\alpha) = \frac{1}{2}w^2 - \alpha(w-1)$ as a curve in w for a few fixed rates. Every curve is a bowl, and every bowl passes through the same point $(1, \frac{1}{2})$, because at $w = 1$ the penalty is zero whatever the rate.



Ask two questions of this picture.

Question A, bowl by bowl: where is the bottom of each bowl? Fix α, look along w, take the lowest point. Those are the black dots: 0, 0.375, 0.5, 0.375. Each bottom is $D(\alpha)$. The *largest bottom* is 0.5, on the $\alpha = 1$ bowl. In symbols this is $\max_\alpha \min_w \mathcal{L}$: first the min along w inside each bowl, then the max across bowls. It is the **dual value** d^*.

Question B, position by position: what is the highest curve above each w? Fix w, look up through all the bowls, take the highest. To the right of $w = 1$ the highest curve is the dashed one,

$\alpha = 0$, so the height is $\frac{1}{2}w^2$. To the left of $w = 1$ the curves climb without limit as α grows, so the height is $+\infty$. Now take the *lowest of these heights* over all w: it is 0.5, at $w = 1$. In symbols this is $\min_w \max_\alpha \mathcal{L}$: first the max across bowls at each position, then the min along w. It is the **primal value** p^*, because "the highest curve above w" came out as $\frac{1}{2}w^2$ on the allowed side and $+\infty$ on the forbidden side, which is exactly the original constrained problem.

Why every bottom is at or below every top. Pick any bowl, say $\alpha = 0.5$, and any position, say $w = 1.5$. The bowl's bottom is at or below the bowl's own height at $w = 1.5$, because the bottom is its lowest point. That height is at or below the highest curve at $w = 1.5$, because that one is the highest. So

$$\text{bottom}(\alpha = 0.5) \;\le\; \mathcal{L}(1.5,\, 0.5) \;\le\; \text{top}(w = 1.5).$$

Nothing depended on the choices, so every bottom is at or below every top, and therefore the largest bottom is at or below the smallest top:

$$d^* = \max_\alpha \min_w \mathcal{L} \;\le\; \min_w \max_\alpha \mathcal{L} = p^* \qquad \text{(weak duality)}.$$

One shared point on one curve is the whole proof.

The duality gap is $p^* - d^*$: smallest top minus largest bottom. On the picture the smallest top is the crossing at height 0.5, and the largest bottom is the black dot on the $\alpha = 1$ bowl, also at height 0.5, and they are the same point. Gap zero. For this to happen some bowl must have its bottom exactly at the crossing, and the $\alpha = 1$ bowl does. This is what **strong duality** means geometrically: one bowl's bottom sits on the crossing point. The table in Section 8 lists the gap for other rates: 0.125 at $\alpha = 0.5$ and at $\alpha = 1.5$, whose bottoms sit below the crossing.

When is the gap zero? For a convex objective with linear constraints and at least one point that satisfies every constraint strictly, always. (E) qualifies, and so does the SVM: $\frac{1}{2}\|\mathbf{w}\|^2$ is a bowl, the constraints are linear, and any separating line scaled up is strictly feasible.

When is it not? When the objective is not a bowl. Minimise $-w^2$ subject to $-1 \le w \le 1$: the answer is $p^* = -1$. Now each curve $\mathcal{L} = -w^2 - \alpha(1 - w) - \beta(1 + w)$ is an upside-down parabola with no bottom, so every "bottom" is $-\infty$ and $d^* = -\infty$. The bound is still true and completely useless. Convexity is what guarantees a curve whose bottom touches the crossing.

For the SVM the picture is the same with more dimensions: one bowl per choice of the eight rates, its bottom is $D(\boldsymbol{\alpha})$, and the crossing is the primal answer $\frac{1}{4}$. The rates $\alpha_1 = \alpha_5 = \frac{1}{4}$ give the bowl whose bottom touches the crossing.

9′. The same five steps on the SVM

Step 1: one rate per constraint. Give constraint i a rate $\alpha_i \ge 0$:

$$\mathcal{L}(\mathbf{w}, b, \boldsymbol{\alpha}) = \tfrac{1}{2}\|\mathbf{w}\|^2 - \sum_{i=1}^{8} \alpha_i \big[\, y_i(\mathbf{w}^\top \mathbf{x}_i + b) - 1 \,\big].$$

The bracket is the *slack* of constraint i: positive when the point is safely outside the corridor, zero on its edge, negative inside. Safe points earn a refund, points inside pay.

Step 2: minimise over $(\mathbf{w}, b)$ by calculus. For fixed rates $\mathcal{L}$ is a bowl in $(\mathbf{w}, b)$:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{w}} = \mathbf{w} - \sum_i \alpha_i y_i \mathbf{x}_i = 0 \;\Rightarrow\; \boxed{\mathbf{w} = \sum_i \alpha_i y_i \mathbf{x}_i} \qquad \frac{\partial \mathcal{L}}{\partial b} = -\sum_i \alpha_i y_i = 0 \;\Rightarrow\; \boxed{\sum_i \alpha_i y_i = 0}$$

The first box is a *recipe* for $\mathbf{w}$ in terms of the rates, not yet a number. (The second box is a condition on the rates: if it fails, $\mathcal{L}$ has no bottom in b, the minimum is $-\infty$, and such rates give a useless bound; that is why the dual carries it as a constraint.)

Substitute both boxes into $\mathcal{L}$. Split it into four pieces, $A = \frac{1}{2}\|\mathbf{w}\|^2$, $B = \sum_i \alpha_i y_i \, \mathbf{w}^\top \mathbf{x}_i$, $C = b \sum_i \alpha_i y_i$, $D = \sum_i \alpha_i$, with $\mathcal{L} = A - B - C + D$. The second box kills C. The first box makes $B = \mathbf{w}^\top \mathbf{w} =$

$\|\mathbf{w}\|^2 = 2A$. So $\mathcal{L} = A - 2A + D = D - A$:

$$D(\boldsymbol{\alpha}) = \sum_i \alpha_i - \tfrac{1}{2}\Big\|\sum_i \alpha_i y_i \mathbf{x}_i\Big\|^2 = \sum_i \alpha_i - \tfrac{1}{2}\sum_{i,j} \alpha_i \alpha_j y_i y_j \, \mathbf{x}_i^\top \mathbf{x}_j.$$

Step 3: $D(\boldsymbol{\alpha}) \le p^*$. Same argument. $D(\boldsymbol{\alpha})$ is the smallest entry of $\mathcal{L}$ over all $(\mathbf{w}, b)$, and the true answer $(\mathbf{w}^*, b^*)$ is one entry. At the true answer every slack is ≥ 0 and every rate ≥ 0, so the penalty is a refund or zero, and $\mathcal{L}(\mathbf{w}^*, b^*, \boldsymbol{\alpha}) \le \tfrac{1}{2}\|\mathbf{w}^*\|^2 = p^*$. The smallest entry is at most this one, so $D(\boldsymbol{\alpha}) \le p^*$.

Steps 4 and 5: the dual.

$$d^* = \max_{\boldsymbol{\alpha}} \sum_i \alpha_i - \tfrac{1}{2}\sum_{i,j} \alpha_i \alpha_j y_i y_j \, \mathbf{x}_i^\top \mathbf{x}_j \quad \text{subject to} \quad \alpha_i \ge 0, \quad \sum_i \alpha_i y_i = 0.$$

Bowl, linear constraints, strictly feasible point: the gap is zero, $d^* = p^*$. One rate per data point instead of one weight per feature, and the data appear only as inner products $\mathbf{x}_i^\top \mathbf{x}_j$.

10. Solving the dual on the eight points

Put a rate only on points 1 and 5: $\alpha_1 = \alpha_5 = \alpha$, all others 0. The condition $\sum_i \alpha_i y_i = \alpha - \alpha = 0$ holds. The recipe gives $\mathbf{w} = \alpha(3,3) - \alpha(1,1) = (2\alpha, 2\alpha)$, and the bound is $D = 2\alpha - \tfrac{1}{2}(8\alpha^2) = 2\alpha - 4\alpha^2$.

rate α	recipe $\mathbf{w}$	bound $D(\alpha) = 2\alpha - 4\alpha^2$	gap $p^* - D(\alpha)$, $p^* = 0.25$
0.1	$(0.2, 0.2)$	0.16	0.09
0.2	$(0.4, 0.4)$	0.24	0.01
0.25	$(0.5, 0.5)$	**0.25**	**0**
0.3	$(0.6, 0.6)$	0.24	0.01

The largest bound: $dD/d\alpha = 2 - 8\alpha = 0$, so $\alpha = \tfrac{1}{4}$, $D = \tfrac{1}{4}$. Recover the line from the recipe: $\mathbf{w} = \tfrac{1}{4}(3,3) - \tfrac{1}{4}(1,1) = (\tfrac{1}{2}, \tfrac{1}{2})$; point 1 has rate > 0 so its constraint is tight, $\mathbf{w}^\top \mathbf{x}_1 + b = 1$, giving $b = 1 - 3 = -2$. Primal value $\tfrac{1}{2} \cdot \tfrac{1}{2} = \tfrac{1}{4}$: $d^* = p^*$, gap zero. (In general two support vectors $\mathbf{x}_+$ and $\mathbf{x}_-$ give $\alpha = 2/\|\mathbf{x}_+ - \mathbf{x}_-\|^2$; here $2/8 = \tfrac{1}{4}$.)

Why no rate on the other six? At the solution, point 2 at $(4, 3)$ has score 1.5, slack 0.5. Raising α_2 from 0 would lower D (a positive slack times a rate is subtracted), so the largest bound keeps $\alpha_2 = 0$. This is the general pattern, called *complementary slackness*: for every point, either the rate is zero or the slack is zero,

$$\alpha_i \left[\, y_i(\mathbf{w}^\top \mathbf{x}_i + b) - 1 \,\right] = 0.$$

A point with room to spare is charged nothing and contributes nothing to $\mathbf{w}$.

Support vectors. $\mathbf{w} = \sum_i \alpha_i y_i \mathbf{x}_i$ with $\alpha_i > 0$ only for points on the corridor's edge. The line is built from those points alone. Remove any other point and nothing moves; nudge a support vector and the line turns.

Why the dual rather than guessing, as in Section 7: with n points in d dimensions there are 2^n patterns of "which constraints are tight". The dual is a single concave quadratic with one maximum; a solver finds the pattern by itself, landing at $\alpha_i = 0$ on the points that do not matter.

11. Using the line, and the door to the kernel

The score of a new email $\mathbf{x}$ is

$$f(\mathbf{x}) = \mathbf{w}^\top \mathbf{x} + b = \sum_{i \in \mathrm{SV}} \alpha_i y_i \, \mathbf{x}_i^\top \mathbf{x} + b, \qquad \widehat{y} = \operatorname{sign} f(\mathbf{x}).$$

For $\mathbf{x} = (2, 3)$: $f = \tfrac{1}{4}(3,3)\cdot(2,3) - \tfrac{1}{4}(1,1)\cdot(2,3) - 2 = \tfrac{15}{4} - \tfrac{5}{4} - 2 = 0.5$, positive, so "spam". Its distance from the line is $0.5/\|\mathbf{w}\| \approx 0.71$: halfway to the corridor's edge, which is 1.41 away.

Two things to notice, both used in the second half of the lecture. First, f is a signed distance times $\|\mathbf{w}\|$, not a probability; thresholds and costs from the evaluation lecture still apply, a probability needs calibration. Second, the dual and the decision function touch the data *only through inner products* $\mathbf{x}_i^\top \mathbf{x}_j$ and $\mathbf{x}_i^\top \mathbf{x}$. Replace every inner product by a function $K(\mathbf{x}, \mathbf{z})$ that is an inner product in some lifted space, and the same machinery draws a curved boundary without ever building the lift. That is the kernel trick (the kernel section of the slides). Allowing points inside the corridor at a price C per unit of slack is the soft margin (the soft-margin section of the slides); its dual is the same as above with the single extra wall $\alpha_i \le C$. There, compute b from a point with $0 < \alpha_i < C$, which sits exactly on the corridor's edge; a point with $\alpha_i = C$ may be inside the corridor and would give the wrong b.

12. The chain in one paragraph

Rule: farthest from the nearest point. Distance: score over $\|\mathbf{w}\|$. Margin: the smallest $y_i\, d(\mathbf{x}_i)$. Goal: maximise the margin, a max of a min, kinked. Scale: nearest score $= 1$, so margin $= 1/\|\mathbf{w}\|$. Primal: shortest $\mathbf{w}$ with every score ≥ 1, a bowl with constraints. Rates: each constraint becomes a penalty at a fixed rate; minimise by calculus; the result is a lower bound on the answer. Dual: choose the rates that make the bound largest; that maximisation is the dual, and for this convex problem the gap is zero. Solution: $\alpha = \frac{1}{4}$ on the two points that touch the corridor, $\mathbf{w} = (\frac{1}{2}, \frac{1}{2})$, $b = -2$, margin 1.414, value $\frac{1}{4}$ on both sides.